Starting System Diagnosis		
Condition	Possible Cause	Correction
	2. Starting circuit wiring faulty.	2. Refer to Wiring. Test and repair faulty starter feed and/or control circuits as required.
	3. Starter motor faulty.	3. Refer to Starter Motor. Replace faulty starter motor as required.
	4. Engine seized.	4. Refer to 9, Engine Diagnosis. Repair or replace faulty engine as required.
Starter engages, spins out before engine starts.	1. Starter ring gear faulty.	1. Refer to Starter Motor. Remove starter motor to inspect starter ring gear. Replace faulty starter ring gear as required.
	2. Starter motor faulty.	2. Refer to Starter Motor. Replace faulty starter motor as required.
Starter does not disengage.	1. Starter motor improperly installed.	1. Refer to Starter Motor. Tighten starter motor mounting hardware to correct tightness specifications as required.
	2. Starter relay faulty.	2. Refer to Starter Relay. Replace faulty starter relay as required.
	3. Ignition switch faulty.	3. Refer to Ignition Switch and Key Lock Cylinder. Replace faulty ignition switch as required.
	4. Starter motor faulty.	4. Refer to Starter Motor. Replace faulty starter motor as required.

TESTING

Before testing the starting system perform a visual inspection of the starting system components and connections.

COLD CRANKING TEST

Refer to **Starting System** in the index of this service manual for the location of complete starting system wiring diagrams. Before performing this test, be certain that the following procedures are accomplished:

- The battery is fully-charged and load tested. Refer to **Battery Charging** for battery charging procedures. Refer to **Battery** for battery diagnosis and testing procedures, including battery load test procedures.
- Fully engage the parking brake.
- If the vehicle is equipped with an automatic transmission, place the gearshift selector lever in the Park position. If the vehicle is equipped with a manual transmission, place the gearshift selector lever in the Neutral position and block the clutch pedal in the fully depressed position.
- Verify that all lamps and accessories are turned off.
- To prevent the engine from starting, remove the Automatic ShutDown (ASD) relay. The ASD relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the underside of the PDC cover for ASD relay identification and location.

(1) Connect a suitable volt-ampere tester to the battery terminals (Fig. 1). See the instructions provided by the manufacturer of the volt-ampere tester being used.

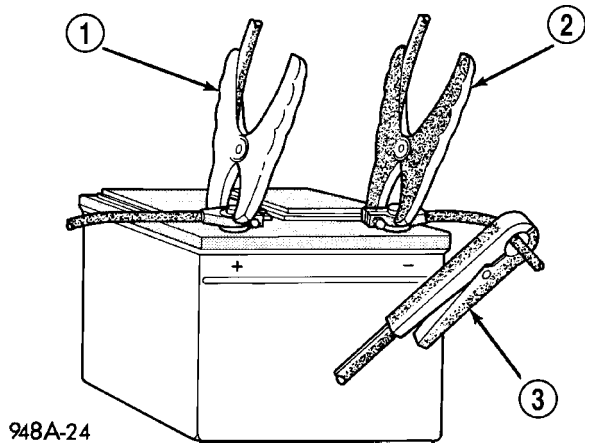

Fig. 1 Volts-Amps Tester Connections - Typical

1 - POSITIVE CLAMP
2 - NEGATIVE CLAMP
3 - INDUCTION AMMETER CLAMP

(2) Rotate and hold the ignition switch in the Start position. Note the cranking voltage and current (amperage) draw readings shown on the volt-ampere tester.

(a) If the voltage reads below 9.6 volts, refer to **Starter Motor** for starter motor diagnosis and testing procedures. If the starter motor tests OK,